

ChE 201 – Introduction to Thermodynamics –Fall 2017

Student Learning Outcome 1

Rubric for assessment problem on final exam (Problem #5)

Application of energy and entropy balances to a adiabatic air compressor

SLO 1 Performance Indicator (PI)	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject Problem 4a, 4b, 4c 15 pts	Does not understand the connection between mathematical models and the system or process to be analyzed or designed (<7 pts)	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development (7-9 pts)	Able to successfully combines mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering (10-15 pts)
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra Problem 4d 20pts	Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems (<7 pts)	Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving (7-11 pts)	Applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems (12-20 pts)
4. Translates academic theory into engineering applications Problem 4e 5 pts	Does not appear to grasp the connection between theory and the problem (0-1 pts)	Some gaps in understanding the application of theory to the problem and expects theory to predict reality (2-3 pts)	Translates academic theory into engineering applications and accepts limitations of mathematical models of physical reality (4-5 pts)

Averages:

Performance Indicators: (1) 2.98 (2) 2.12 (4) 2.06

Overall Average: 2.39

Average scores showed 49/50 students were very strong in formulating the engineering system and model equations. A majority were satisfactory or were developing the skills to solve engineering problems and translating theory into engineering applications (34/50 and 30/50). PI 2 and PI 4 had lower average scores. In order to address these outcomes, more problem solving will be done in class and in the discussion session with the TA. The emphasis will be on how to approach the problem, setting up the engineering model, identifying the relevant equations and performing the calculations. In addition, the physical aspects of the problems will also be identified to connect with engineering applications.

QUESTION 5: (40 pts)

A 2.5 kg/min stream of air is compressed in an adiabatic compressor at steady state from an initial state of 100 kPa and 295K to a final pressure of 900 kPa. Assume the air is an ideal gas, and neglect any changes in kinetic and potential energy.

- a) (5 pts) Draw a schematic of the system and label the system boundary.
- b) (5 pts) State the engineering model.
- c) (5 pts) Write the simplified energy balance and simplified entropy balance.
- d) (20 pts) If the compressor is reversible, determine the outlet temperature and the minimum power required (*show all your work below for full credit*)

$$T_2 = \text{_____ K} \quad \dot{W} = \text{_____ kW}$$

- e) (5 pts) Draw the adiabatic compression process on an **h vs s** diagram.

Solution

CHE 201 Final Exam

December 14th, 2017

QUESTION 5: (40 pts)

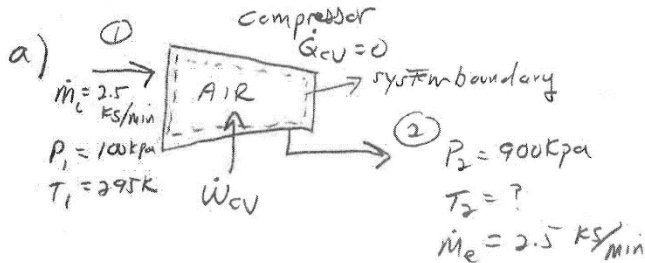
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- (5 pts) State the engineering model.
- (5 pts) Write the simplified energy balance and simplified entropy balance.
- (20 pts) If the compressor is reversible, determine the outlet temperature and the minimum power required (*show all your work below for full credit*)

$$T_2 = 548.8 \text{ K} \quad \dot{W} = -10.76 \text{ kW}$$

(5 pts)

- Draw the adiabatic compression process on an h vs s diagram.



- 1) Air flowing in a compressor is the control volume
- 2) C.V. at steady state, $\frac{dE_{CV}}{dt} = 0$ $\frac{dS_{CV}}{dt} = 0$
- 3) C.V. adiabatic, $\dot{Q}_{CV} = 0$
- 4) Air modeled as an ideal gas
- 5) $\Delta KE, \Delta PE = 0$
- 6) $\dot{W}_{CV}$ is only work mode (shaft work)

c) **Energy Balance:** $\frac{dE_{CV}}{dt} = \dot{Q}_{CV} - \dot{W}_{CV} + \dot{m}(h_1 - h_2) + \Delta KE + \Delta PE$
 $\dot{m}_1 = \dot{m}_2 = \dot{m}$
 $\dot{W}_{CV} = \dot{m}(h_1 - h_2)$

ENTROPY Balance $\frac{dS_{CV}}{dt} = \frac{\dot{Q}_{CV}}{T} + \dot{m}(s_1 - s_2) + \dot{S}_{CV}$
 $0 = \dot{m}(s_1 - s_2) + \dot{S}_{CV}$

d) Compressor is reversible, so $\dot{Q}_{CV} = 0$ and Entropy Balance becomes; $S_1 = S_2$ isentropic process

Determine T_2 : Find $S_1(T_1, P_1) \Rightarrow S_2(T_2, P_2)$

use eqn. 6.20a $S(T_2, P_2) - S(T_1, P_1) = S^\circ(T_2) - S^\circ(T_1) - R \ln \frac{P_2}{P_1}$

since $S_1 = S_2$ Rearrange: $S^\circ(T_2) = S^\circ(T_1) + R \ln \frac{P_2}{P_1}$

From Table A-22 $S^\circ(295K) = 1.68515 \frac{kJ}{K \cdot kg}$

$P_2 = 900 \text{ kPa}$

$P_1 = 100 \text{ kPa}$

$\bar{R} = 8.314 \frac{kJ}{K \cdot mol}$

MW Air = 28.97 $\frac{kg}{Kmol}$

$S^\circ(T_2) = 1.68515 \frac{kJ}{K \cdot kg} + \left(\frac{8.314 \text{ kJ}}{K \cdot mol} \cdot \frac{1 \text{ kmol}}{28.97 \text{ kg}} \right) \times \ln \left(\frac{900 \text{ kPa}}{100 \text{ kPa}} \right)$

$S^\circ(T_2) = 1.68515 + (0.2870)(2.1972) = 1.68515 + 0.63057$

$S^\circ(T_2) = 2.31572 \frac{kJ}{K \cdot kg}$ Find on Table A-22 T_2 need to interpolate between 540 2.29906 and 550 2.31809

$\frac{T_2 - 540}{550 - 540} = \frac{2.31572 - 2.29906}{2.31809 - 2.29906}$

$T_2 = \frac{(0.01666)}{(0.01903)}(10) + 540 = 0.87546(10) + 540$

$T_2 = 548.8 \text{ K}$

d.) Minimum power? at $T_2 = 548.8\text{K}$

1st Law $\dot{W}_{cu} = \dot{m} (h_1 - h_2)$

Table A-22

$h_1(295\text{K}) = 295.17$

$h_2(548.8\text{K})$
interpolate:

$$\frac{h_2 - 544.35}{554.74 - 544.35} = \frac{548.8 - 540}{550 - 540}$$

$$h_2 = \left(\frac{8.8}{10}\right)(10.39) + 544.35 = 9.14 + 544.35 \frac{\text{KJ}}{\text{kg}}$$

$h_2 = 553.49 \frac{\text{KJ}}{\text{kg}}$

$$\begin{aligned} \dot{W}_{cu} &= \left(\frac{2.5 \text{ kg}}{\text{min}}\right) (295.17 - 553.49) \frac{\text{KJ}}{\text{kg}} = (2.5)(-258.32) \\ &= -645.8 \frac{\text{KJ}}{\text{min}} \left| \frac{1 \text{ min}}{60 \text{ sec}} \right| \frac{1 \text{ kW}}{1 \text{ KJ/s}} = \boxed{-10.76 \text{ kW}} \end{aligned}$$

